

PharmaFind AI System Architecture

PRR/ROR pharmacovigilance, agentic tool-calling, PubMed RAG, PMID guardrails, and PDF reporting.

1. Interaction layer - deterministic dashboard and agentic experience

Deterministic review dashboard

Configurable settings, signal table, evidence drawer, Markdown preview, and PDF download.

Agentic chat workflow

Natural-language review, compare, explain-signal, and generate-PDF requests. The agent calls backend tools instead of inventing results.

Qualified reviewer

Uses outputs as evidence support for human review, not as diagnosis, incidence, or causality.

2. Backend orchestration - typed API, bounded agent tools, and session cache

FastAPI router

Typed endpoints: /reviews, /reviews/pdf, /agent/chat, and health checks.

Review toolset

Runs FAERS counts, PRR/ROR statistics, PubMed RAG retrieval, and report rendering.

Agent runner

Bounded tool loop for review, compare drugs, explain selected signal, and render cached PDF.

Session cache

Stores last drug, summary, Markdown report, PDF path, and follow-up context.

3. Deterministic FAERS signal engine - transparent statistics are the source of truth

openFDA counts

Complete-window aggregate report totals: N, drug count, event count, and drug-event count.

2x2 table

Derives a, b, c, and d at report level for every drug-event pair.

PRR and ROR

Calculates disproportionality measures from transparent formulas.

95% intervals

Adds confidence intervals and zero-cell continuity correction when needed.

Thresholds

Applies minimum cases, minimum PRR, and lower ROR confidence bound.

Flagged signals

Ranks reporting patterns for qualified human review only.

4. PubMed RAG evidence engine - runs per flagged signal and cites retrieved PMIDs only

Drug aliases

Uses brand, generic, active-substance, and FAERS-reported names for searches.

PubMed retrieval

NCBI ESearch and EFetch retrieve titles, abstracts, metadata, DOI, and PMIDs.

RAG context

Only retrieved papers become available context for ranking and synthesis.

Rerank optional

OpenAI embeddings can reorder abstracts by semantic relevance.

Grounded synthesis

Optional LLM summarizes only retrieved abstracts with structured output.

PMID validation

Any cited PMID must exist in the retrieved allow-list or it is rejected.

5. Reviewer outputs and safety boundaries

Dashboard artifacts

Signal table, evidence drawer, citations, and settings snapshot.

Markdown and PDF

ReportLab renders the current Markdown brief without rerunning analysis.

Agentic answer

Chat response can compare drugs, explain signals, and link cached PDF output.

Safety framing

Reporting signal only. Not medical advice, not incidence, not causal proof.

Sequence note: the dashboard and agent both call the same backend toolset. Flagged FAERS signals enter the PubMed RAG lane for retrieved-abstract grounding and PMID validation.